



MEHRAN UNIVERSITY OF ENGINEERING AND TECHNOLOGY, JAMSHORO
DEPARTMENT OF ELECTRONIC ENGINEERING
INTEGRATED ELECTRONICS



Roll No:	Date:
Checked by:	Score:

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT	PRESENTATION	CALCULATION AND CODING	OBSERVATION/ RESULTS	SCORE

Performance Objective:

- Design and Analyze Multi-input CMOS NAND /NOR Gate (3 or 4 Inputs)
- Design Motor Speed Controller by PWM using 555 timer with Astable Mode

Apparatus:

I. Design and analyze CMOS Multi Input NAND Gate

- 4-PMOS Transistors (2N6851)
- 4-NMOS Transistors (2N6659)
- Digital Switches (2)
- VDD (5V)
- GND
- Probe

II. Design Motor Speed Controller by PWM using 555 timer Astable Mode

- 555 timer IC
- Diodes (3)
- Potentiometer 10k (2)
- Resistors
- Capacitors
- Motor
- Mosfet
- Battery (24V)
- Power Supply and GND
- Oscilloscope

Theoretical Background:

I. Design and analyze CMOS Multi Input NAND Gate:

Complementary Metal Oxide Semiconductor abbreviated as “CMOS is a logic family in which uses complementary and symmetrical pairs of p-type and n-type IGFET’s (MOSFET) transistors for logic functions to be performed.

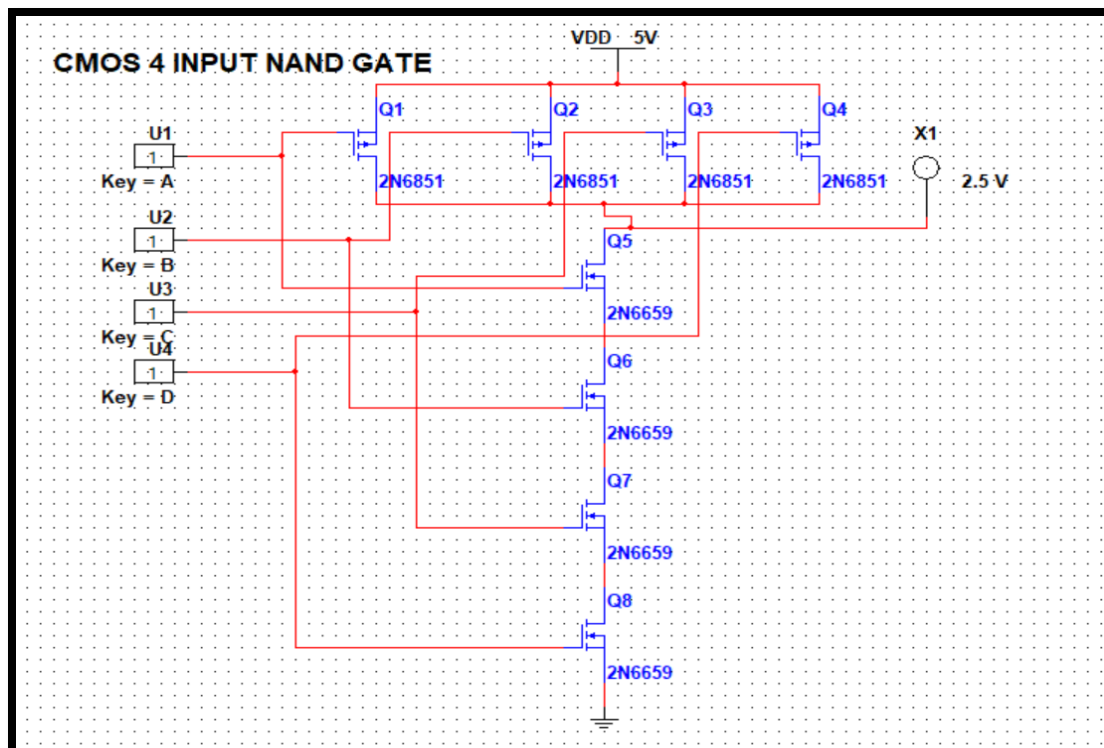
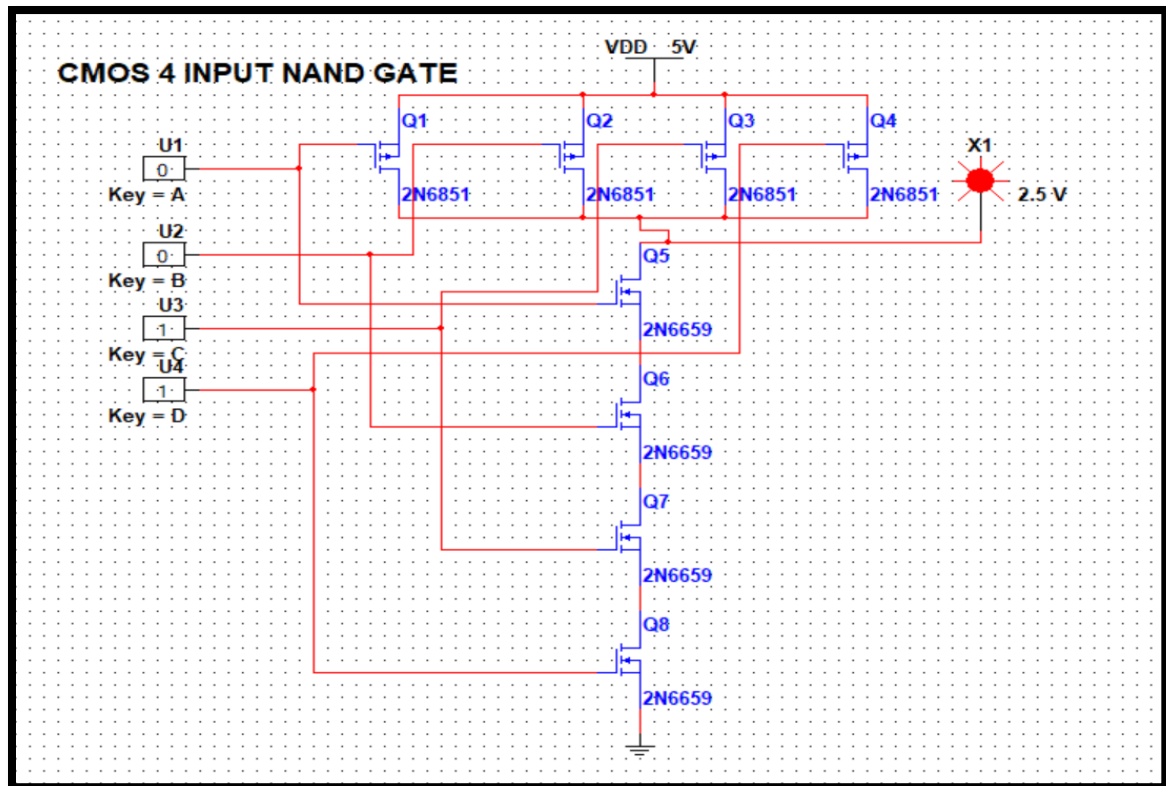
A static CMOS gate is a combination of two networks, called the pull-up network (PUN) and the pull-down network (PDN). The function of the PUN is to provide a connection between the output and VDD anytime the output of the logic gate is meant to be 1 (based on the inputs). Similarly, the function of the PDN is to connect the output to VSS when the output of the logic gate is meant to be 0.

The PUN and PDN networks are constructed in a mutually exclusive fashion such that one and only one of the networks is conducting in steady state. In this way, once the transients have settled, a path always exists between VDD and the output F, realizing a high output (“one”), or, alternatively, between VSS and F for a low output (“zero”). This is equivalent to stating that the output node is always a low-impedance node in steady state.

For designing of basic gates (NAND) the basic functionality of both gates is mentioned.

- A NAND gate is a digital circuit which gives HIGH output only when both the inputs are LOW and if all of the input is HIGH then it produces LOW output.

Circuit

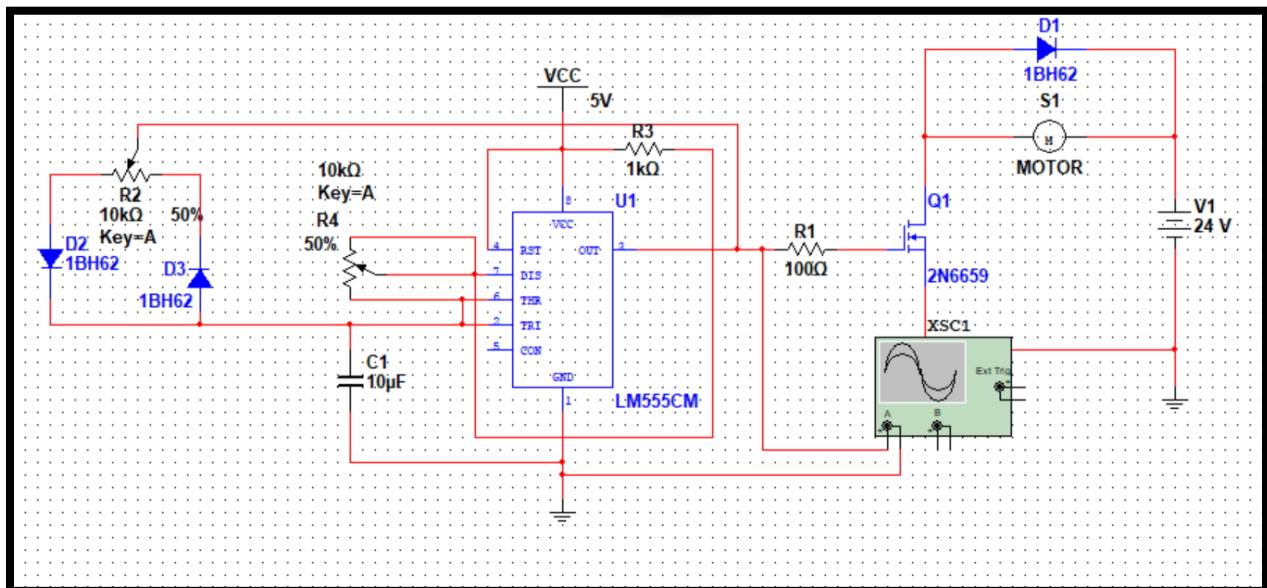


II. Design Motor Speed Controller by PWM using 555 timer Astable Mode

DC motor converts electrical energy in the form of Direct Current into mechanical energy i.e., rotational movement of the motor shaft. The working principle of DC motor is that when a current carrying conductor is placed in a magnetic field, it experiences a mechanical force and starts rotating. Its direction of rotation depends upon Fleming's Left-Hand Rule. The direction of rotation of the shaft of the motor can be reversed by reversing the direction of Direct Current through the motor. The motor can be rotated at a certain speed by applying a fixed voltage to it. If the voltage varies, the speed of the motor varies. Thus, the DC motor speed can be controlled by applying varying DC voltage, whereas the direction of rotation of the motor can be changed by reversing the direction of current through it. For applying varying voltage, we can make use of the PWM technique. For reversing the current, we can make use of H-Bridge circuit or motor driver ICs that employ the H-Bridge technique or other any other mechanisms. DC motors are used in many applications like robot for movement control, toys, quad copters, CD/DVD disk drive in PCs/Laptops etc.

A PWM circuit based on timer NE555 is the heart of this circuit. NE555 is wired as an astable multivibrator whose duty cycle can be adjusted by varying the POT R1. The output of IC1 is coupled to the base of transistor Q1 which drives the motor according to the PWM signal available at its base. Higher the duty cycle the average voltage across motor will be high which results in higher motor speed and vice versa.

Circuit:



Output Waveforms when 2 potentiometers middle terminal is varied

